

Simulation 2

James Ryan, ECE448 Power Electronics, Spring 2026

q1

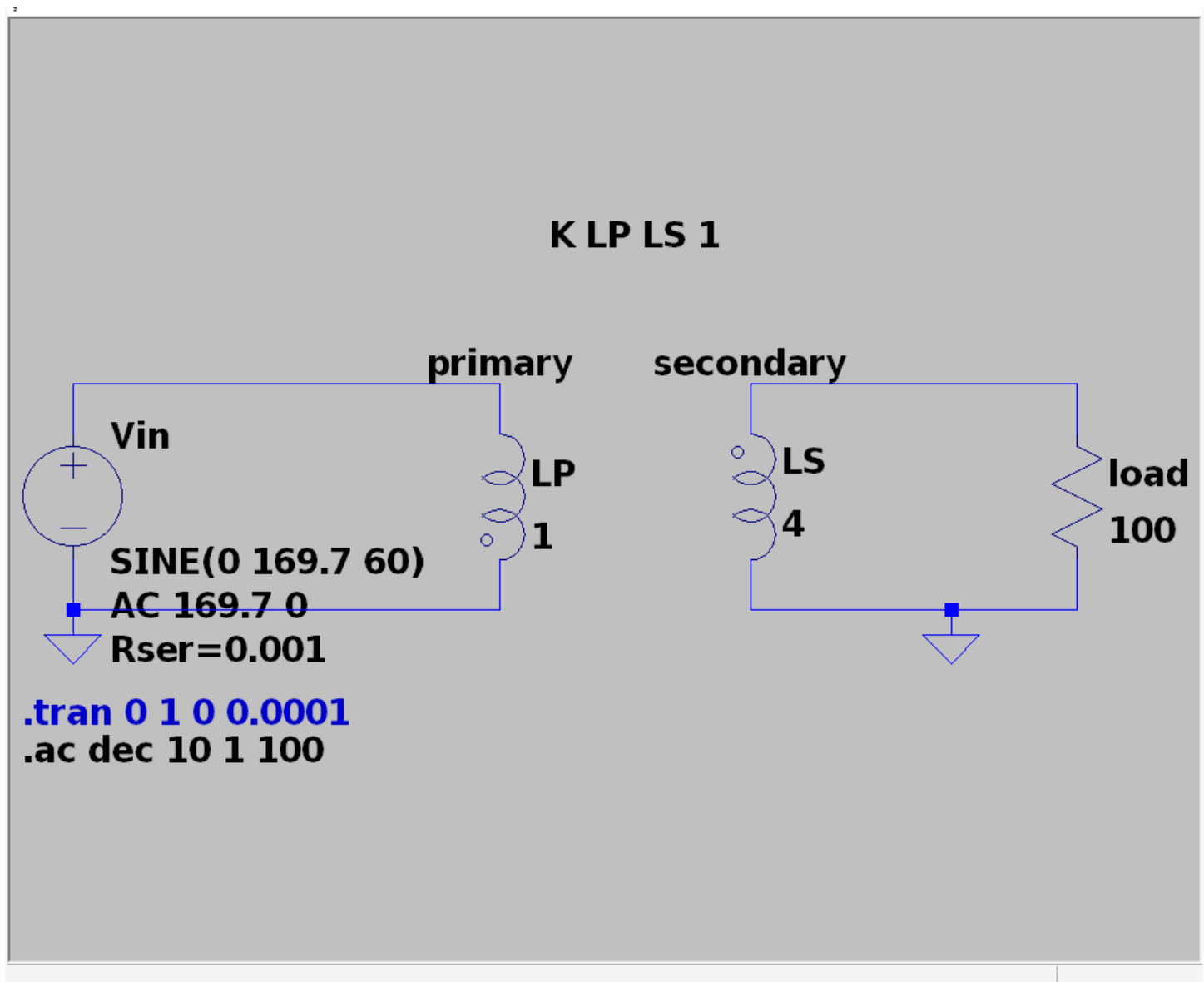


Figure 1: Starting circuit

a

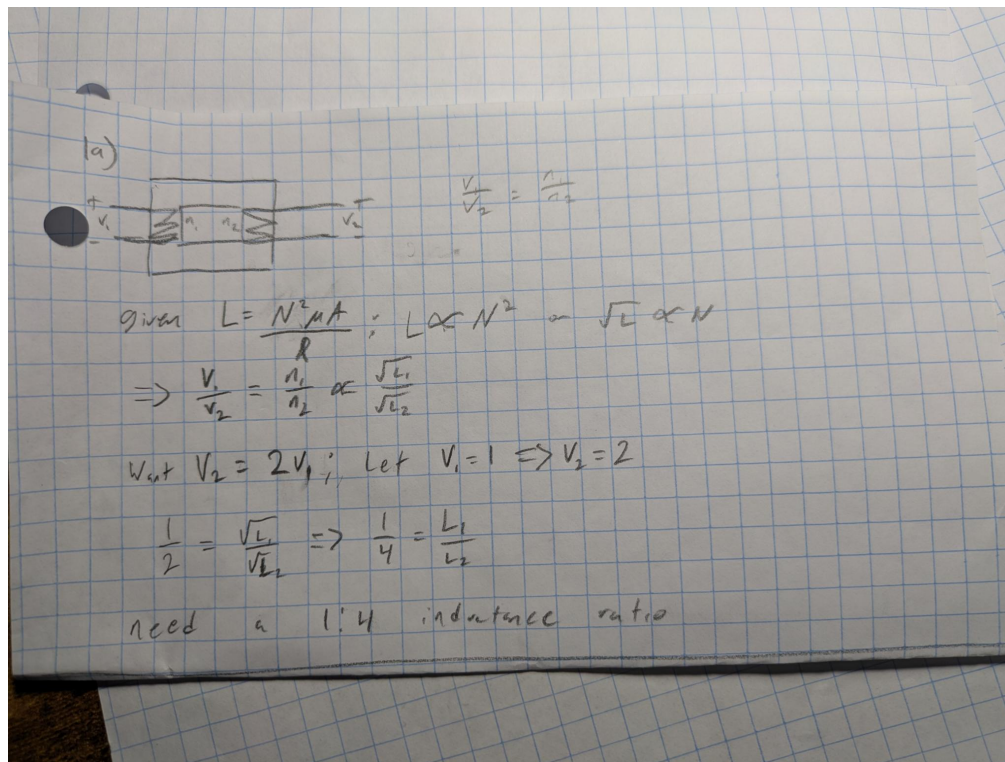


Figure 2: Part a calculations

b

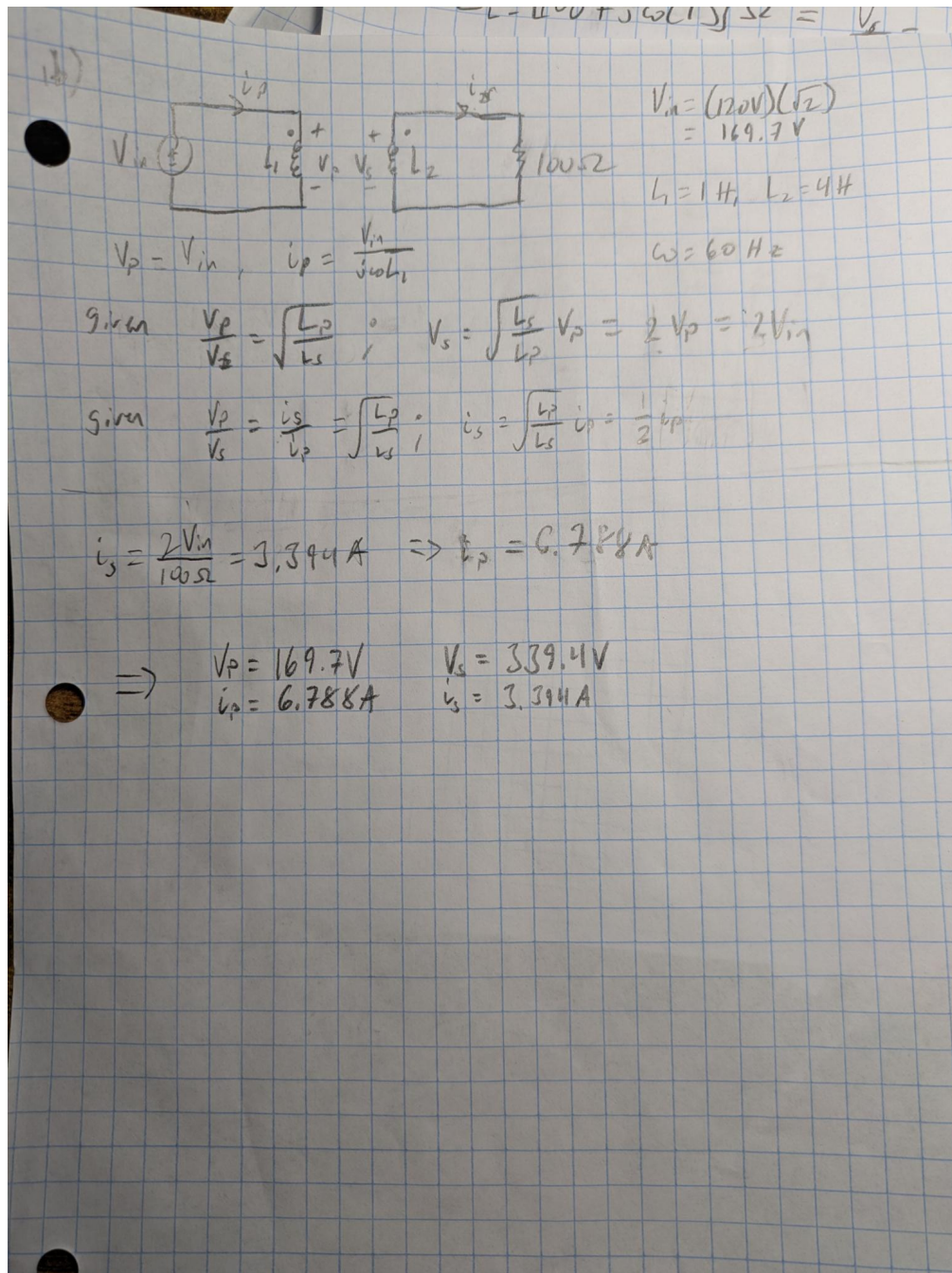


Figure 3: Part b calculations

Note: We are operating on peak magnitudes here, not RMS.

c

While maintaining the turn ratio between the primary and secondary coils, I found that advancing the primary coil to $1e-6$ henries caused the current in the secondary coil to top out at a peak amplitude of 1.20 amperes.

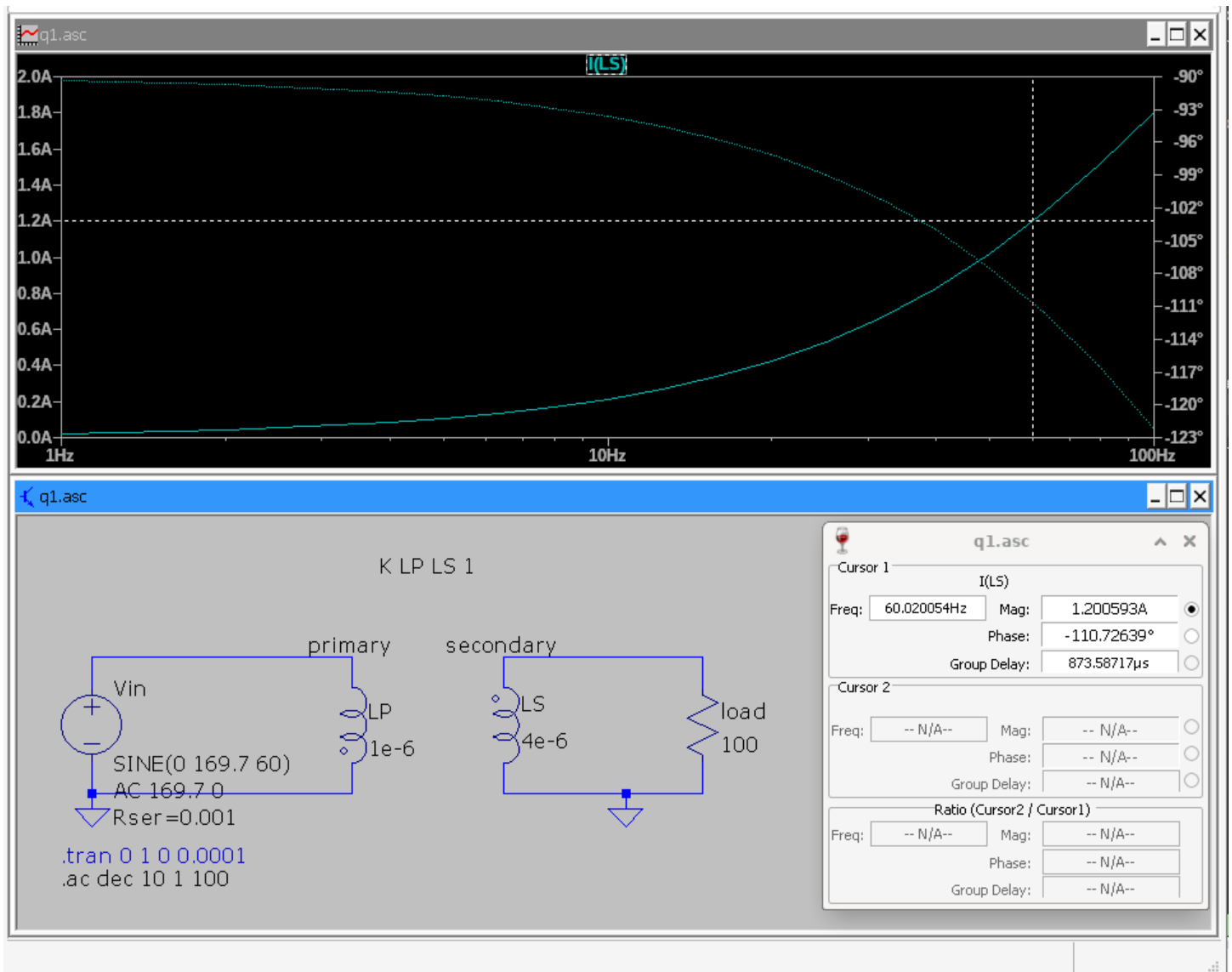


Figure 4: Secondary current after setting the primary coil to $1e-6$ henries

Stepping back a decade gave a result closer to what was calculated at the previous part.

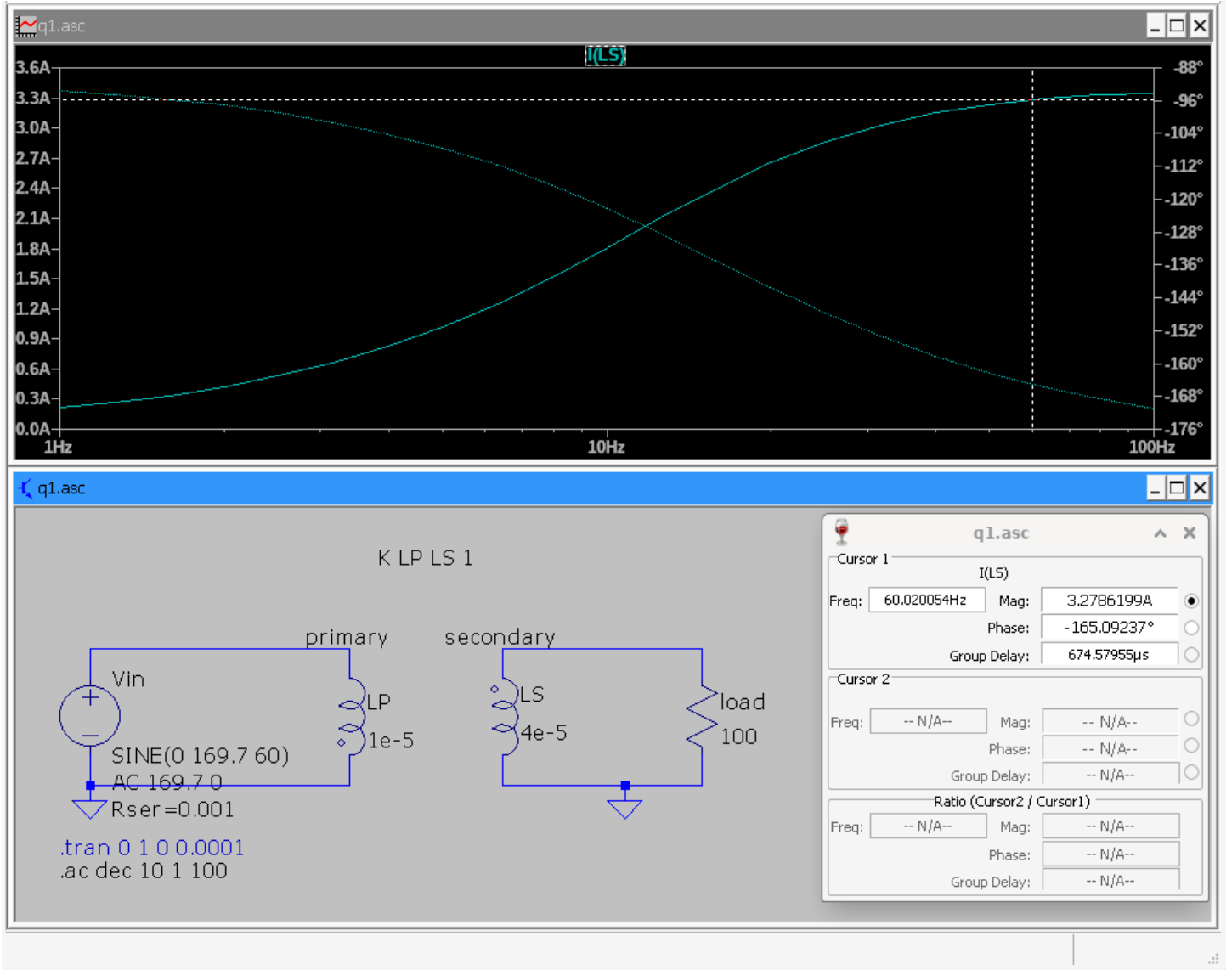


Figure 5: Secondary current after setting the primary coil to $1e-5$ henries

The primary coil's current approaches infinity as we drag the inductance down because the impedance of an inductor is governed by $j\omega H$. Given that our ω is a constant 60 Hz, we can say that by dropping our H we are dropping the series impedance, allowing more current to flow from the power source.

It should be noted that our secondary coil's current is being limited by this drop. Consider the relationship between a primary and secondary coil: $\frac{v_p}{v_s} = \frac{i_s}{i_p}$. Our v_p is constrained as a constant peak voltage of 169 volts, and our secondary voltage v_s has to obey $v_s = 2v_p$ due to our inductance ratio. Therefore, in the relationship between current and voltage between the primary and secondary coils, we have to obey $i_s = \frac{v_p}{v_s} i_p = \frac{1}{2} i_p$, so our i_s will decrease to maintain the enforced ratio.

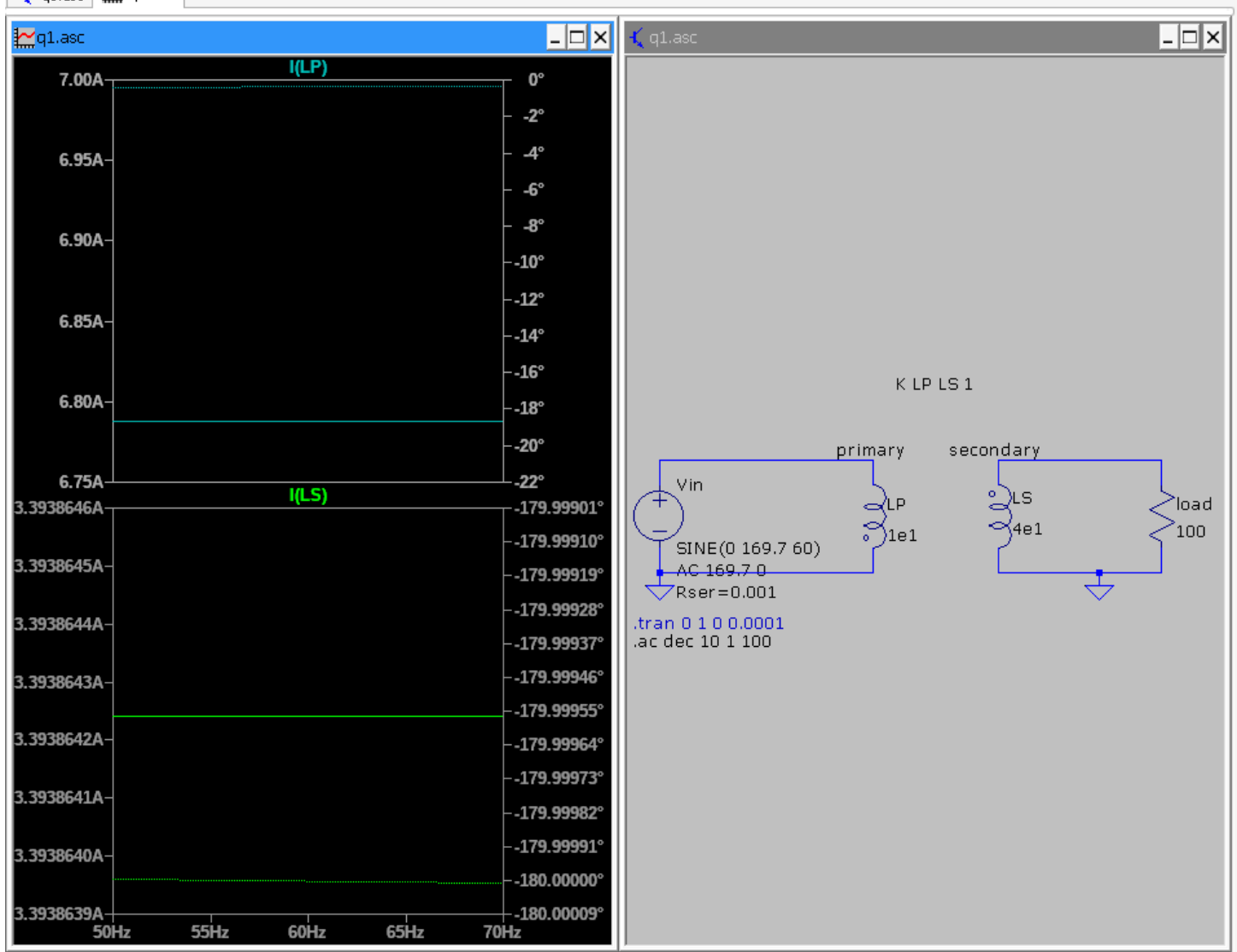


Figure 6: Stepping the coils up to 10:40 gave nice results for our current

When the inductors were raised to 10H and 40H, we can see that we now are observing the expected current magnitudes which were calculated in part b.

d

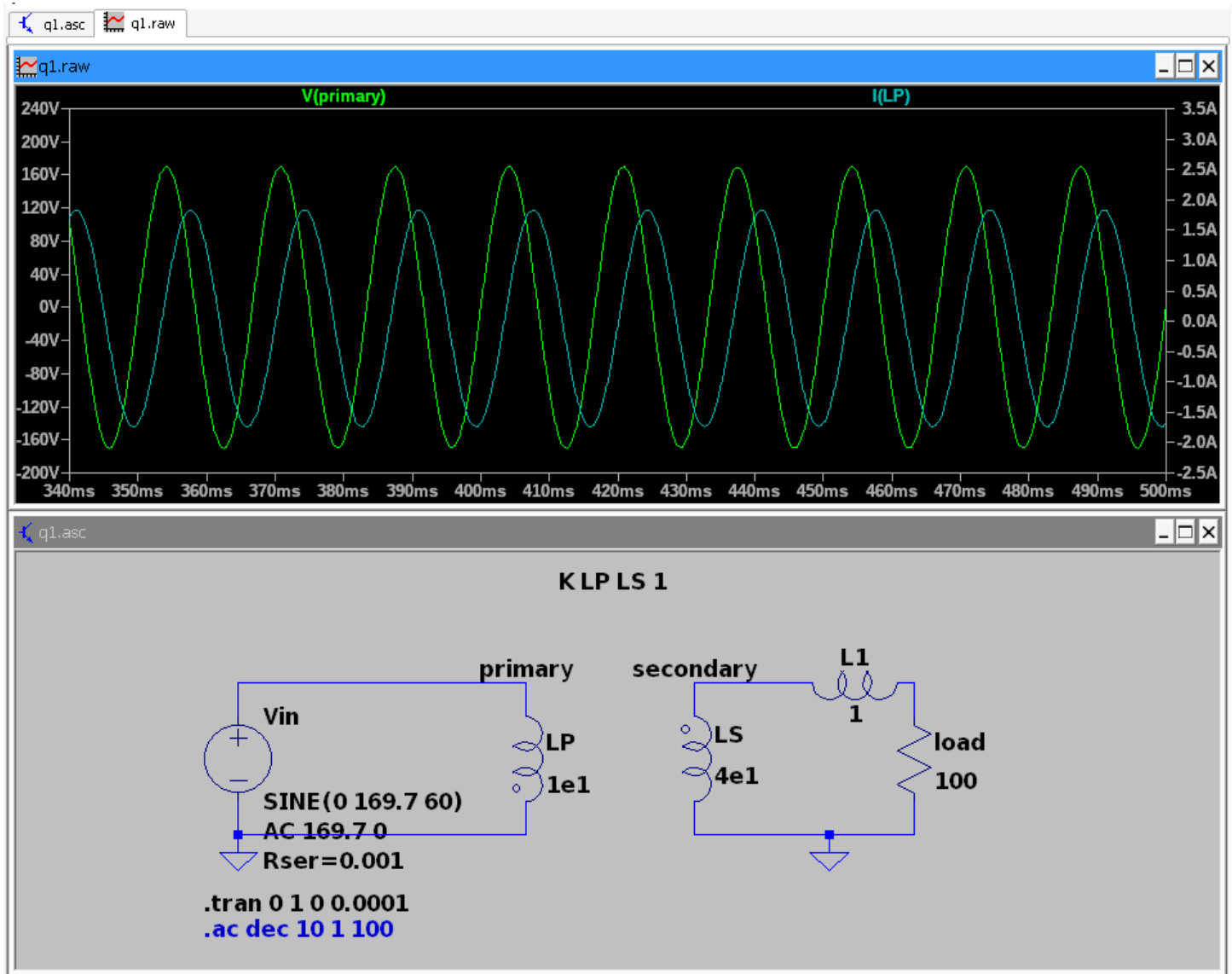


Figure 7: LR motor replaces the secondary load.

Voltage leads current. So, power factor is lagging (inductive).

e

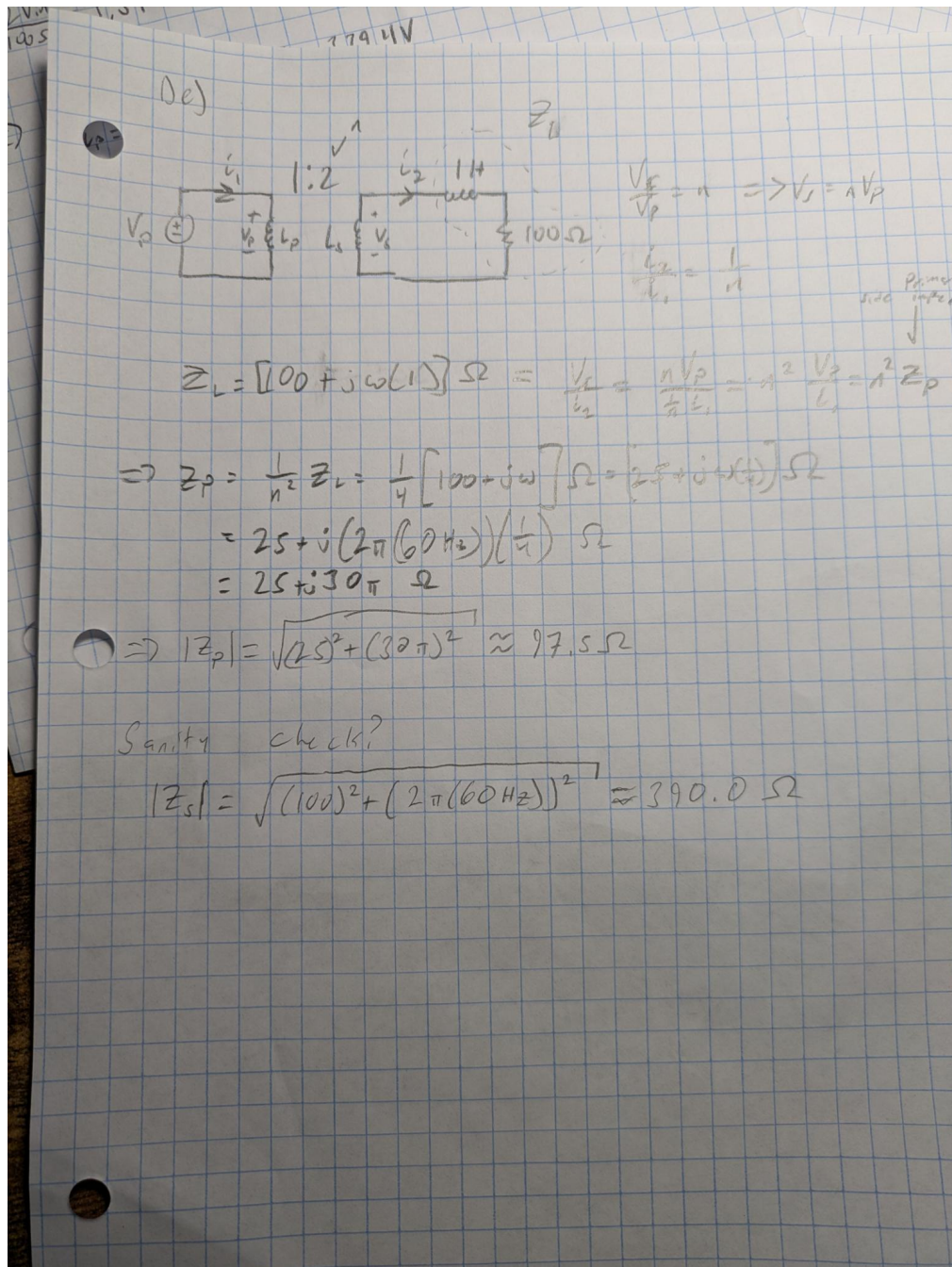


Figure 8: Reflected impedance calculations

f

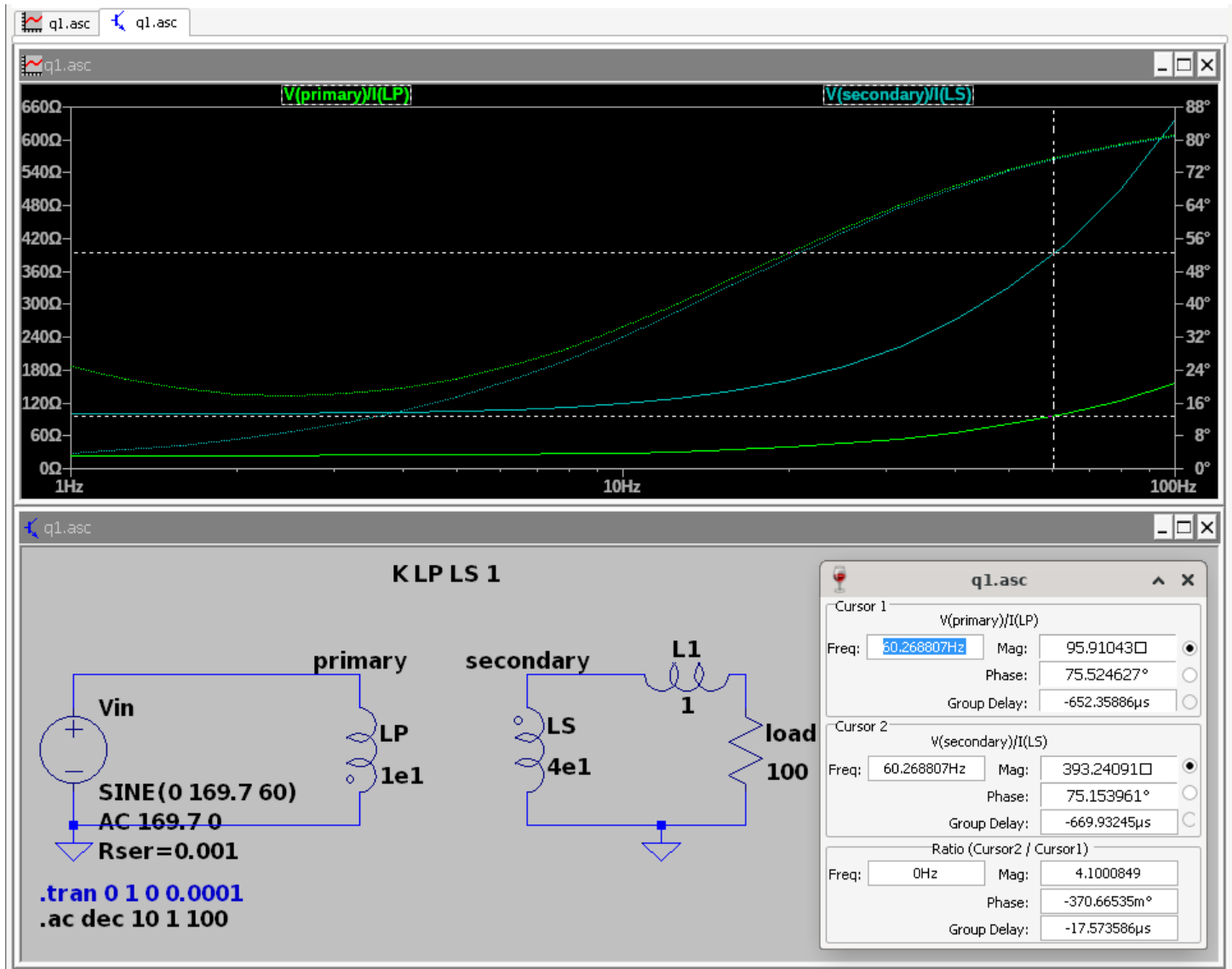


Figure 9: Impedance on primary and secondary sides. Green: Primary, Teal: Secondary

At $f = 60$ Hz, we can see that the primary side sees approximately 95Ω , which is close to our calculated magnitude (on cursor 1).

As a sanity check...

I see the correct load impedance on the secondary side. Treating LS as a source, I expect to see $\sqrt{(100)^2 + (120\pi)^2} \approx 390\Omega$ on the other side, which is there (see the blue trace, cursor 2).

g

I would want to change the K directive, namely i'd drop 1 to a value lower than 1 and at least 0.

According to the LTspice Docs the K directive models the mutual inductance between two inductor objects, so setting this to a value less than 1 would indicate a flux leakage not contributing to the power transfer.

q2

Center-tap transformer

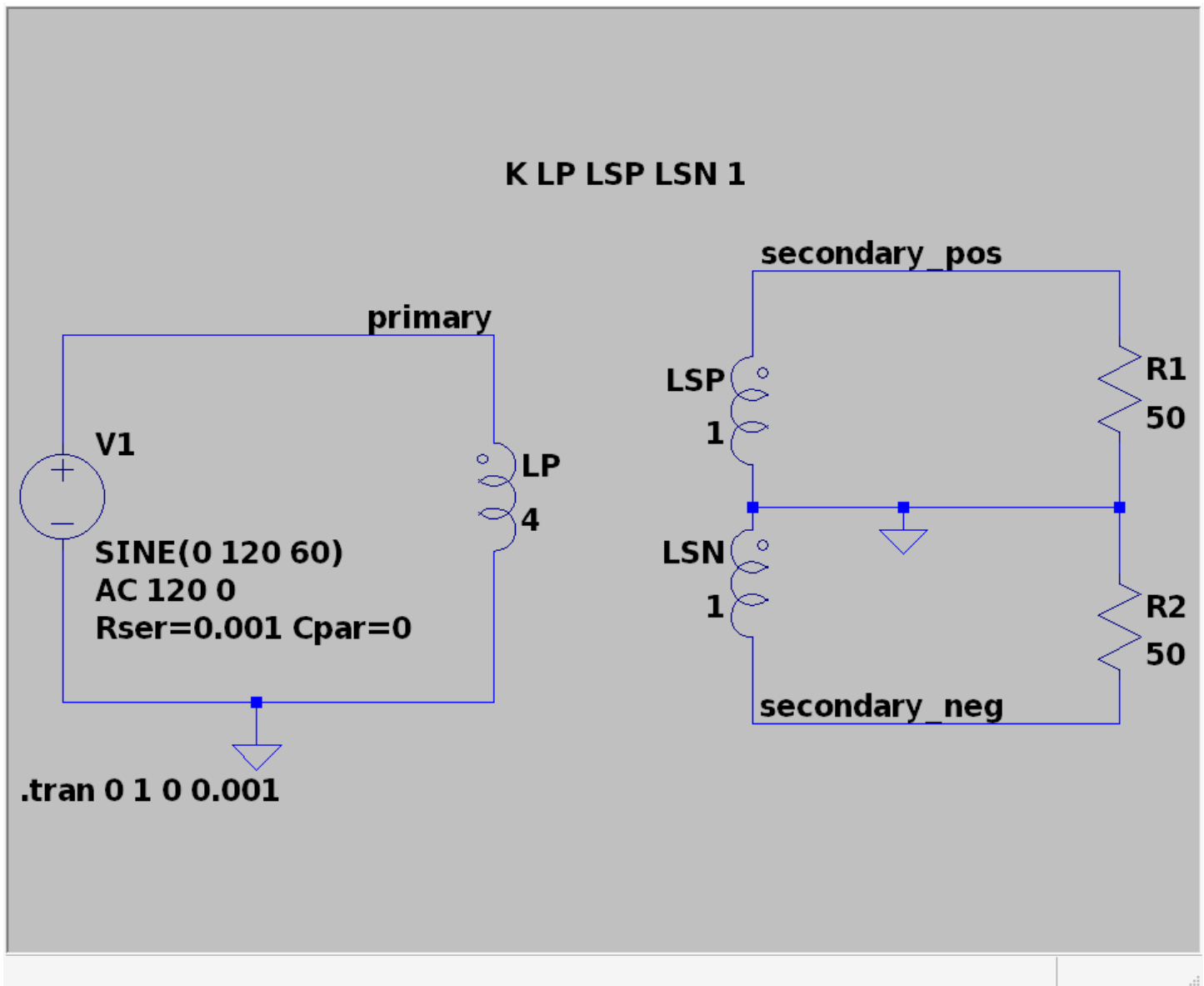


Figure 10: Center-tap base transformer

Note: I assume by setting the correct dot convention, we should follow what you have on the slides for this question. This does make sense to me, since it works out that the top node is in-phase, and the bottom node is out-of-phase for this.

The primary purpose of a center tap transformer is to tap the center of the transformer to a fixed reference voltage. In this case, we are fixing it to ground. So, we can produce two differential waveforms from a fixed reference wave.

In our case, the primary and secondary (tapped) inductor have an equal number of windings, but our secondary is fixing our voltage at the center to ground. This effectively creates two differential waves, each at half the V_{peak} of our primary input.

The easiest way to explain this is thinking of the isolating transformer. For an isolating transformer, the voltage difference seen at the primary should be equal to the voltage difference seen at the secondary. Introducing the center tap enforces the center of the transformer to be at ground, creating two transformers with half the windings relative to the primary. Therefore, our “positive” side will rise in phase with the primary. However, we have the “negative” side below the ground reference, which is also half the windings. ***The voltage difference between the primary coil and the tapped secondary must be equal.*** So, our negative side is pulled down, creating a differential, 180 degree out of phase waveform, also at half peak relative to the primary.

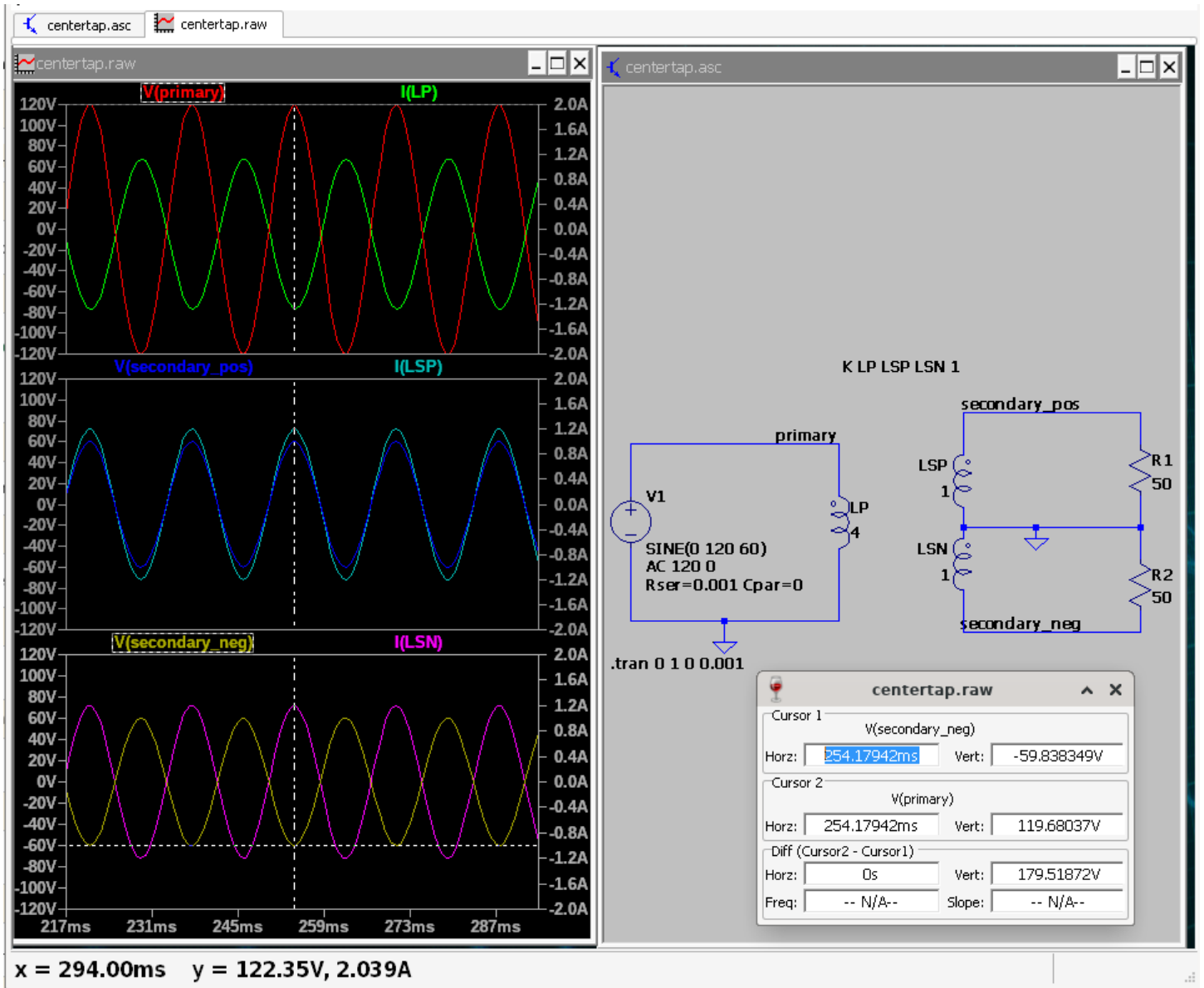


Figure 11: Transient plots of the primary coil, and each side of the center-tapped secondary coil

From this plot, it can be said:

On the primary side, the current is almost 180 degrees out of phase with the voltage, since the primary side is transferring its energy to the secondary coils. In this case, when considering the voltage lead and lag of the primary coil, we should flip the polarity of the current, or shift it backwards by 180 degrees.

This makes intuitive sense from the power triangle. If we measure the current in the opposing direction, we see a 180 degree phase shift back. We're allowed to do this since we are thinking in terms of providing current, which our primary coil is doing for our secondary coils.

We can measure the change in $\theta_{i \rightarrow v}$. Looking at the AC plot should let us calculate this. We'll use a similar method for the later transformers, too.

I derived a lot of my logic from this [stackoverflow post](#)

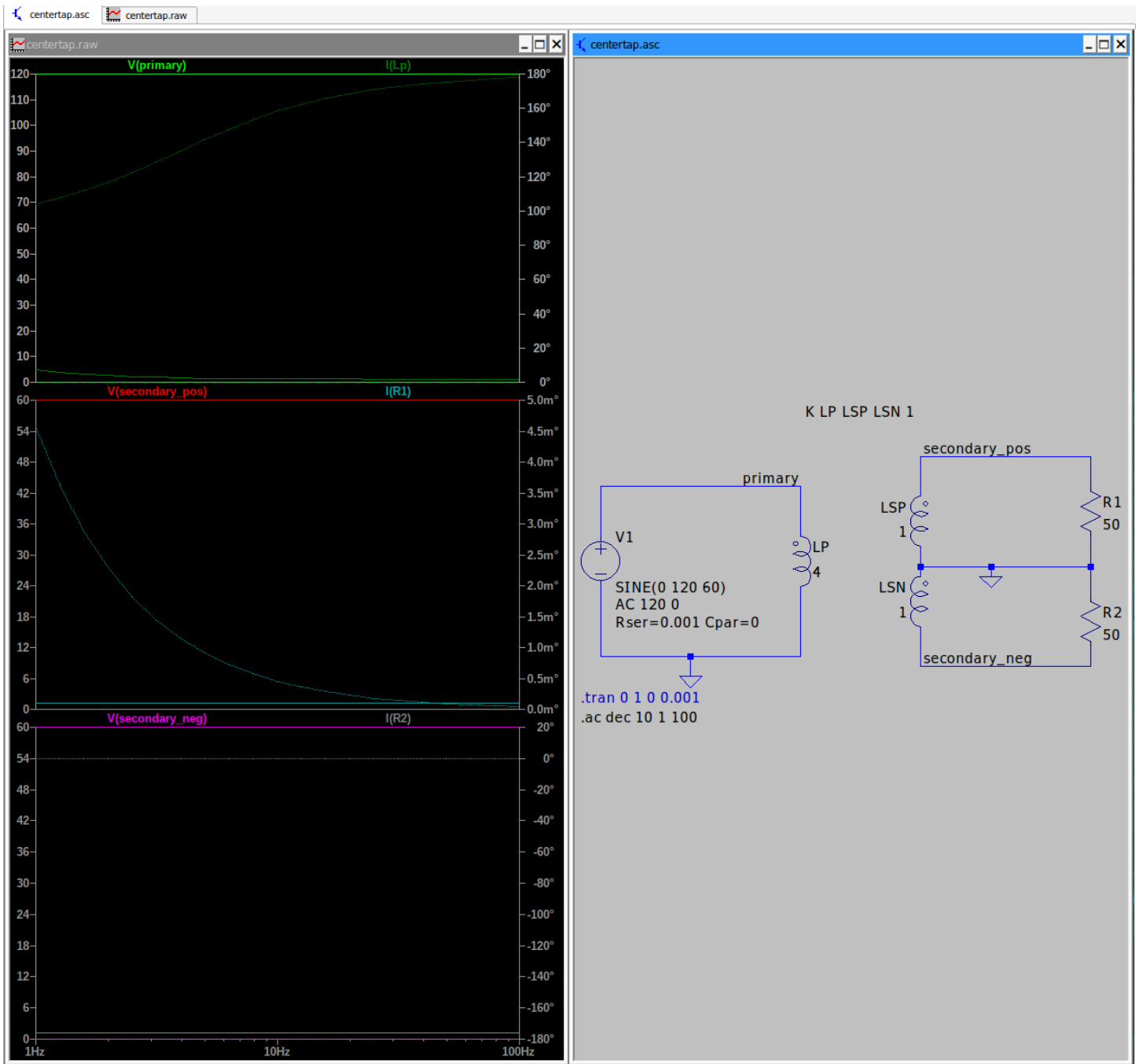


Figure 12: AC plots of the primary coil, and each side of the center-tapped secondary coil

From the transient results:

The **current** waveform on the primary side is *out of phase* with the secondary side's current. The coils on the secondary side, expectedly, have their current waveforms in phase, because they're wired in series. This is expected, thinking of Lenz's law and Faraday's Law: the direction of the induced EMF will oppose the direction of the change in flux. So, the direction of our current in the primary coil opposes the direction of the current in the secondary coil, too. (*Aside:* We'll see this pattern appear again in future transformers.)

The **voltage** waveform on the secondary's positive side is *in phase* with the primary, and the **voltage** waveform on the secondary's negative side is *180 degrees out of phase* with the primary. Expected from the center tap!

And, from the AC results:

We can say (given that the primary coil is a source for the secondary coils) that, since the primary current is *less than* 180 degrees, when we do our sign conversion, we can see that the voltage waveform slightly leads the current on the primary side.

There is no lead or lag on the secondary side, its dominated by resistive loads, phasing differences are only caused by voltage sign convention differences.

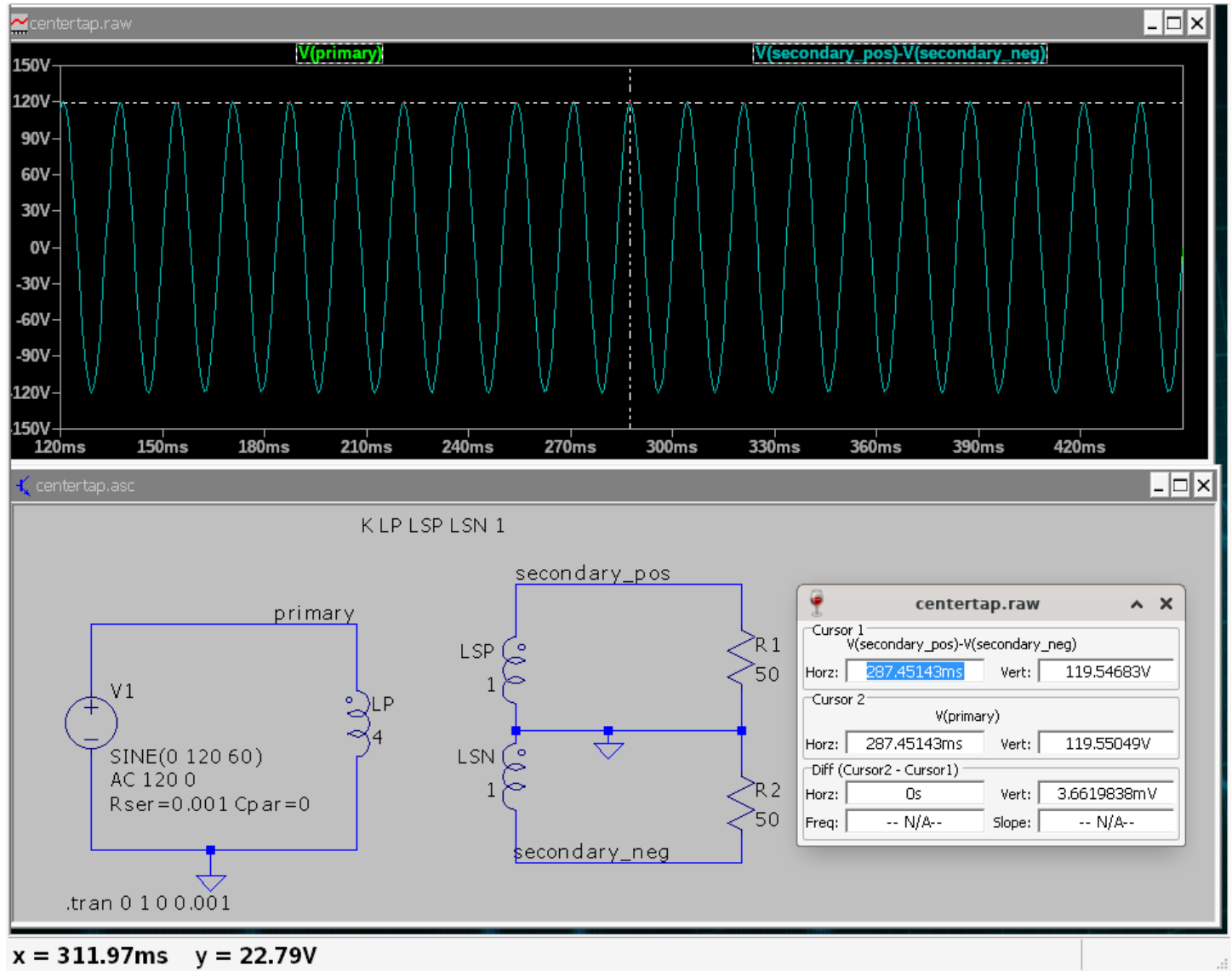


Figure 13: The total voltage drop across the secondary side is equal to the primary side

Balun Transformer

The Balun (*Balanced-Unbalanced*) transformer is configured to have one balanced side, where neither terminal is grounded, and one unbalanced side, where one terminal is held to ground.

Similarly to the center-tapped transformer, the total voltage drop across the primary side should be equal to the voltage drop across the secondary side (NOT considering turn ratio).

Differential mode

This uses a differential input, with $V_{peak} = 120$ V, running at 60 Hz.

We expect to see our output waveform scale to $2V_{peak}$. In this differential setup, the largest drop across the primary coil will be from +120 V to -120 V. Since the secondary side has one net held to ground, our other net will be induced to maintain this drop, and will be pulled to 240 V, and will be in-phase with the signal attached to the phase-dot terminal.

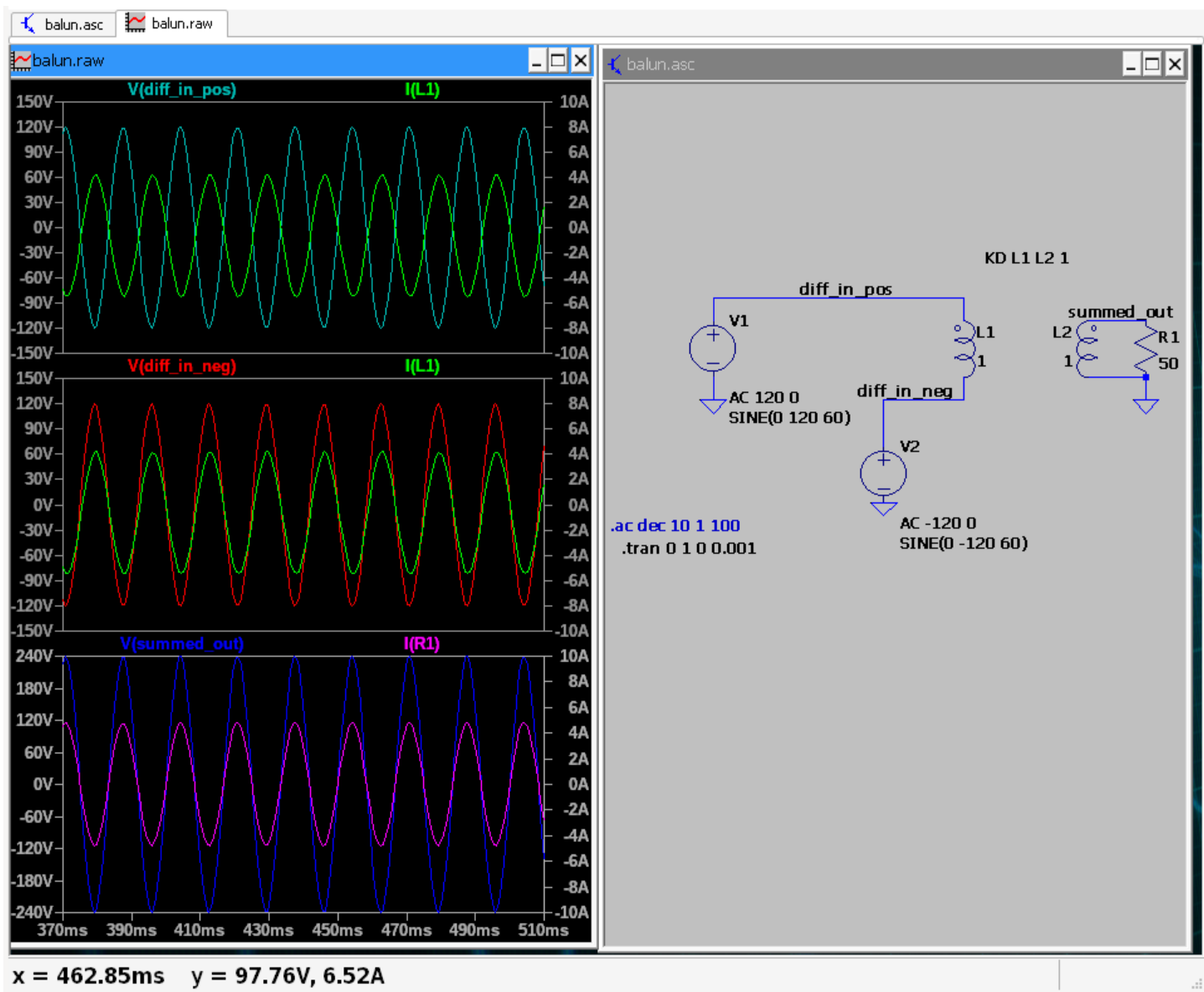


Figure 14: Transient Voltage, Current from the differential inputs, and the added result

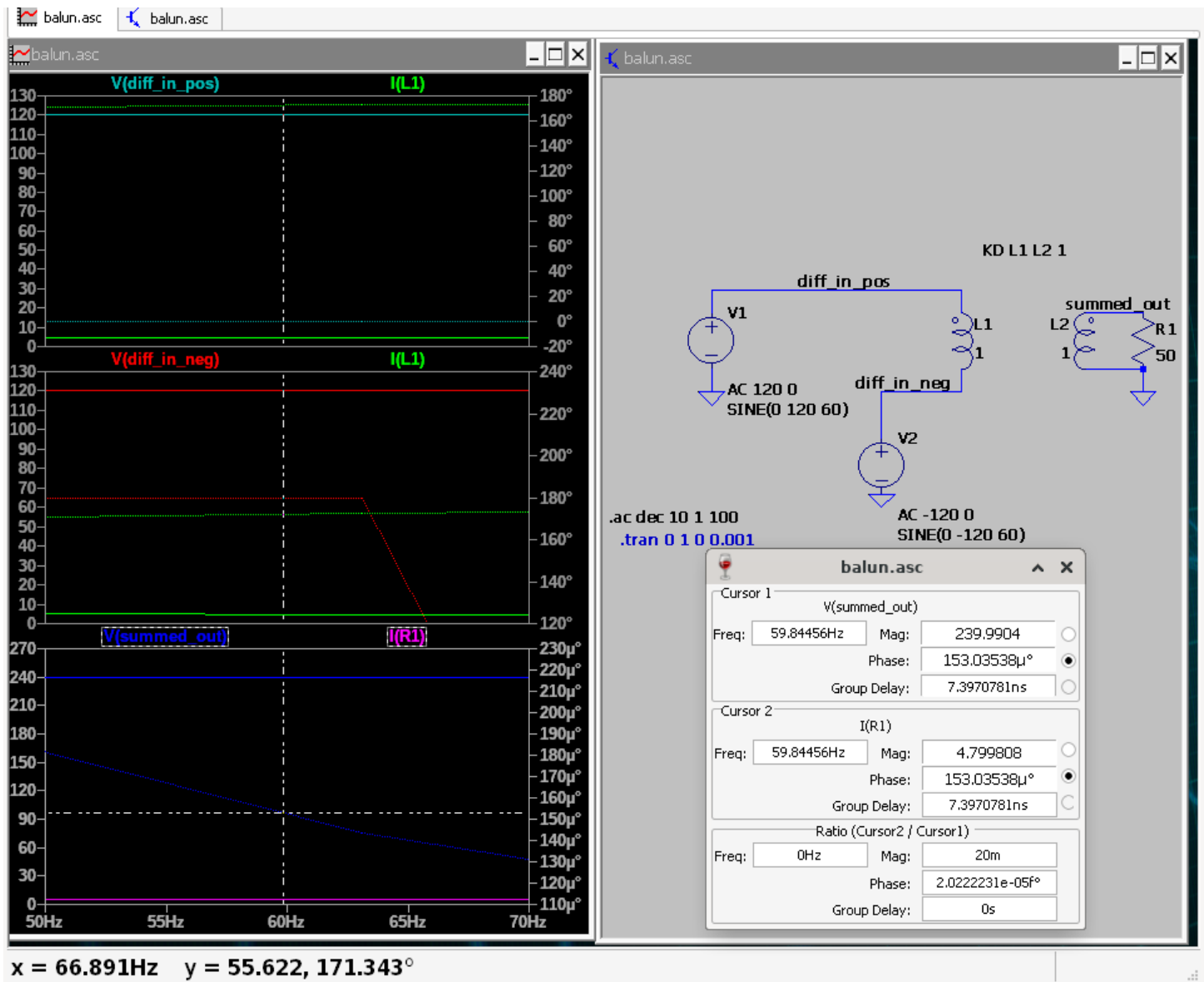


Figure 15: AC Voltage, Current response for differential inputs, result

We can see (especially in the AC results) that there is a slight voltage lead in the primary windings, for similar reasons as described in the center tap.

The current on the sum-side is 180-degrees out of phase with the current on the double-ended side. This is expected, considering Lenz's law and Faraday's law, and our convention considerations.

The summed 240 V result is in-phase with the **pos** differential input. This intuitively makes sense: the induced current is out of phase with the current on the primary side. And, our secondary load is entirely resistive. So, we expect our voltage to be entirely in phase with the current on the secondary side. **THEREFORE!** Our voltage peaks on the secondary side will track the voltage peaks on the primary side, since our current peaks on the secondary side track the current troughs on the secondary side, and we are guaranteed to have voltage out of phase with current on the primary side from Lenz's law.

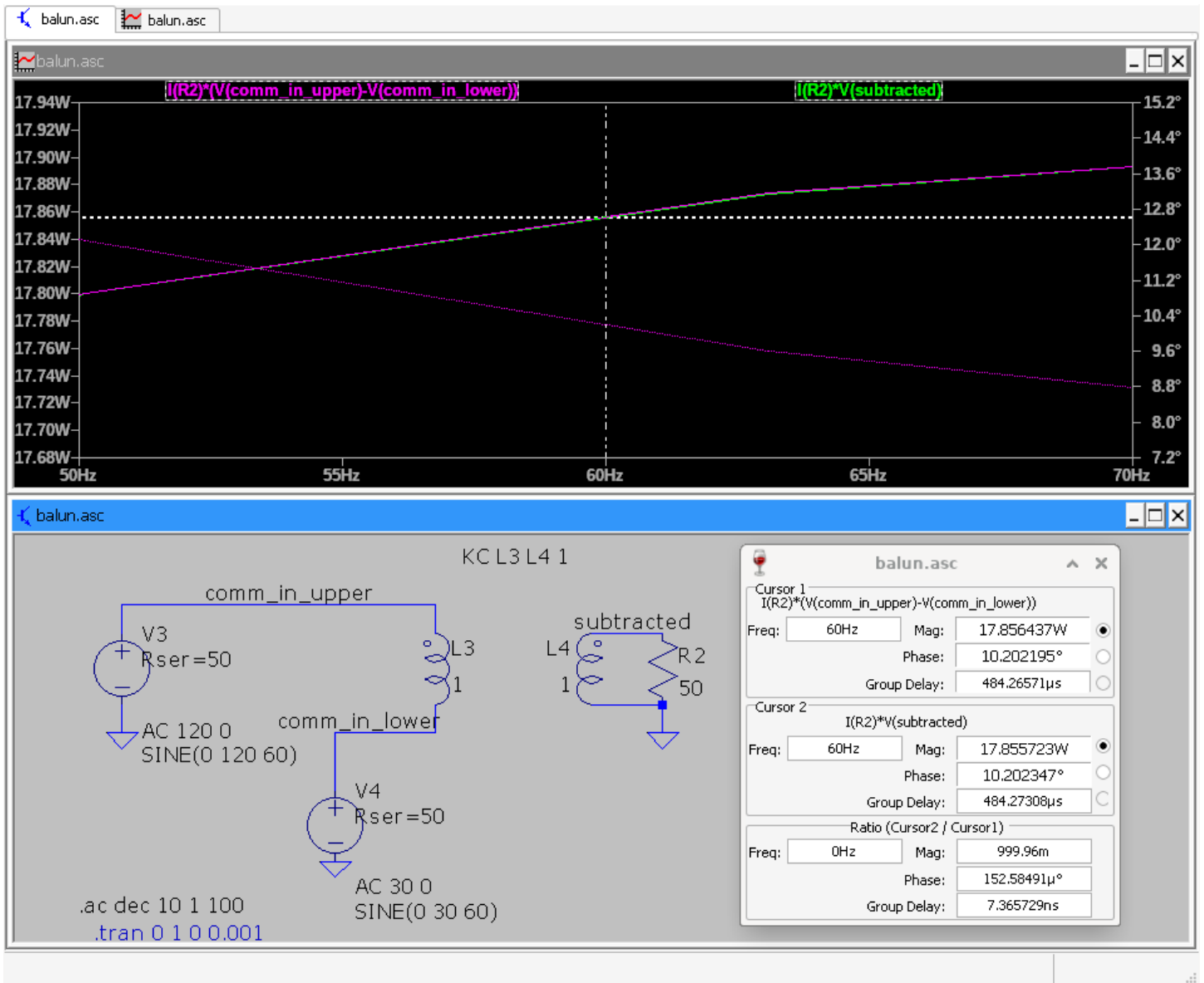


Figure 16: Demonstration of power conservation across the balun

There is about 1 W lost between primary and secondary, presumably due to the slight resistance in L1 and L2.

Common mode

This uses a common mode input, where both `comm_in` nets are running in phase with each other. They do not necessarily have the same V_{peak}

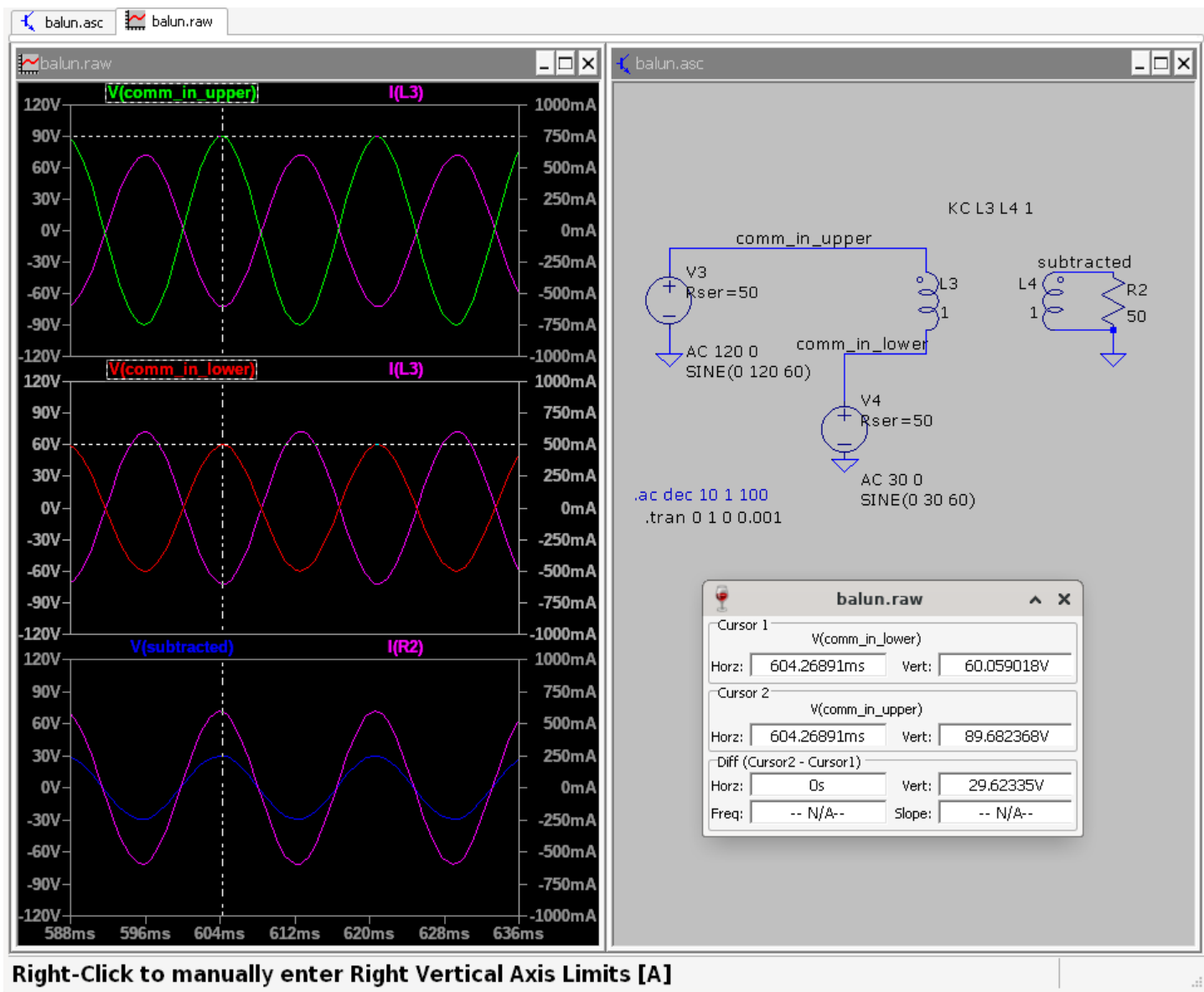


Figure 17: Transient Voltage, Current from each common input, and the subtracted result

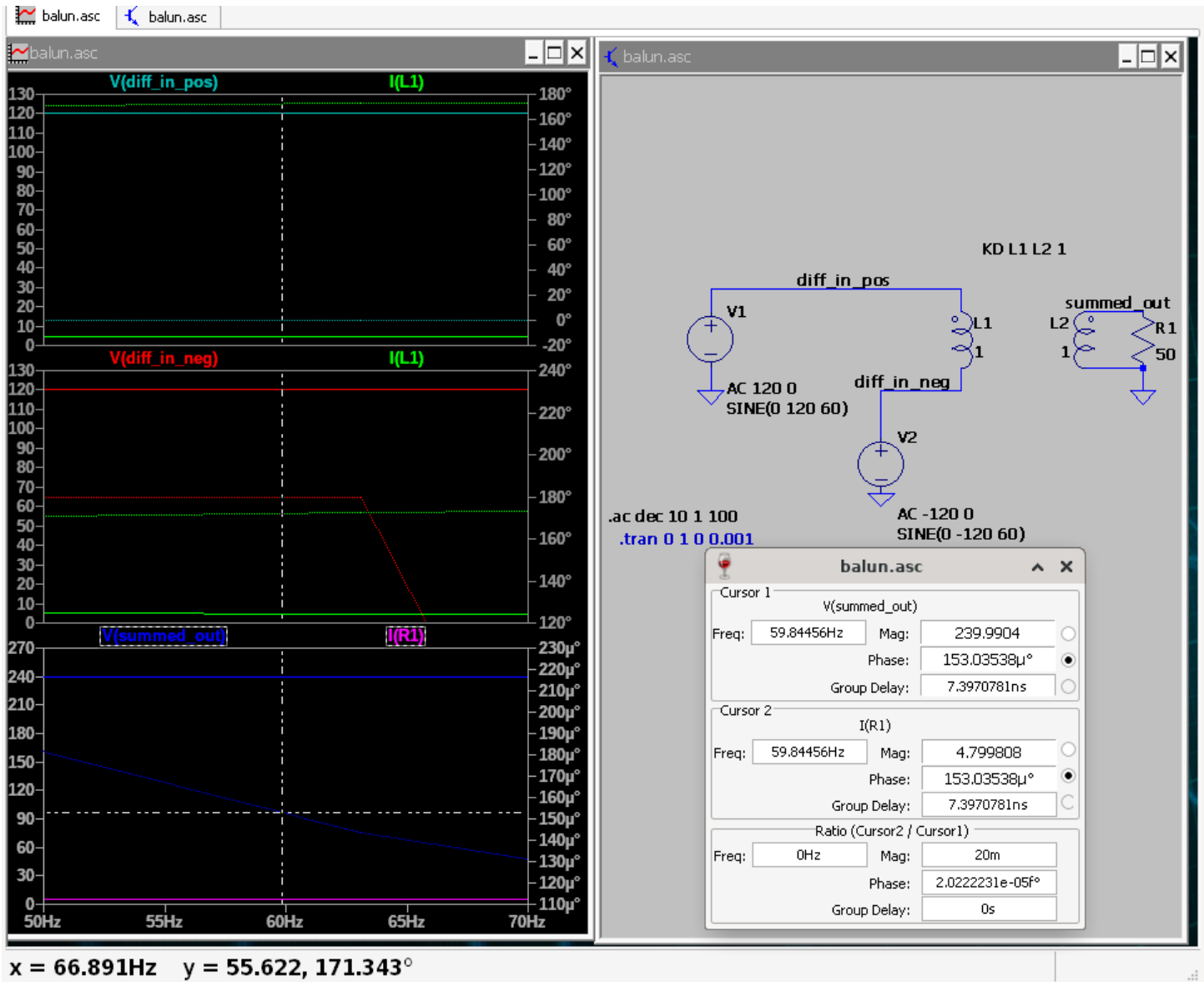


Figure 18: AC Voltage, Current response for common inputs, result

Note that V3 and V4 have $R_{ser}=50$. This was done to center the current waveforms at 0 so it's a little more clear what's happening.

We can see that, at the **upper** net we observe a peak voltage of around 90 V, and on the **lower** net we see 60 V. Therefore, the total drop across the coil should be 30 V, which can be seen on the secondary side.

We see (especially on the AC plot) that both common inputs experience a voltage lead.

We see a similar trend with the voltage and current waveforms: the current on the primary side is out of phase with the current on the secondary side, and the current on the secondary side is in phase with the voltage on the secondary side.

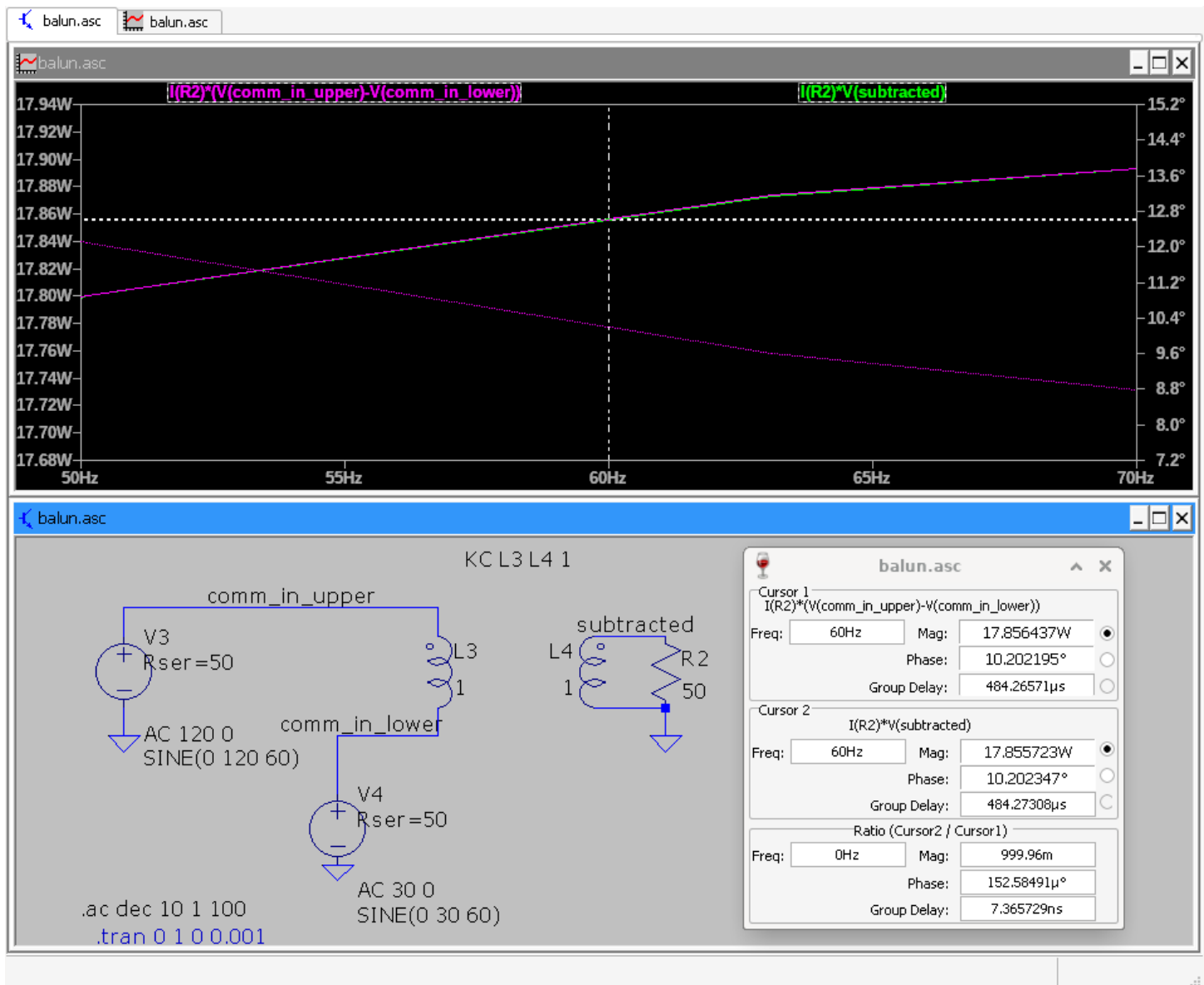


Figure 19: Demonstration of power conservation across the balun

Multi-winding transformer

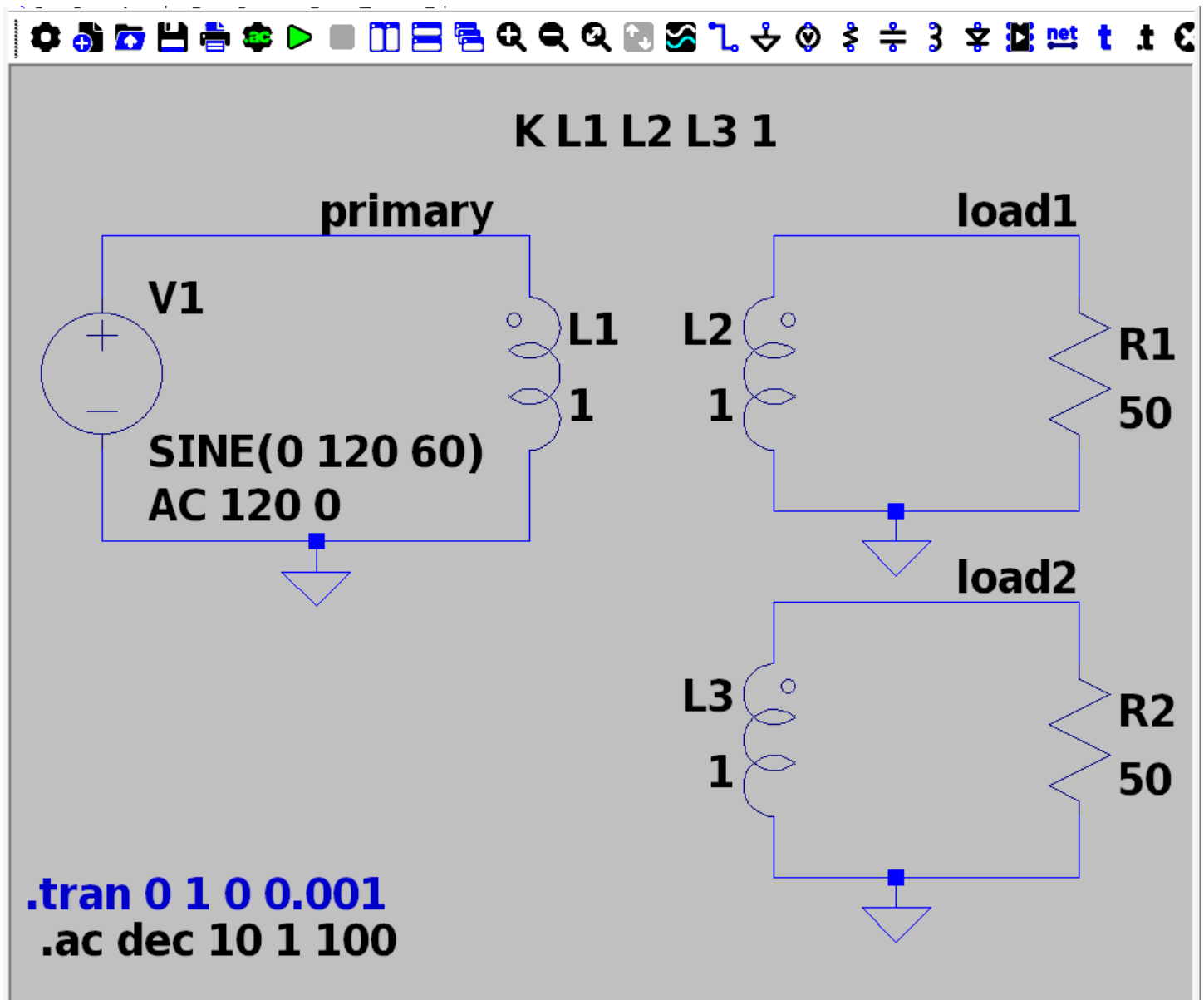


Figure 20: Multi-winding configuration

This is configured to be an isolating transformer; the inductance ratio between the primary coil and any of the load coils is 1:1. Therefore, there should be no change in the magnitude of the voltage when observing on the primary or secondary side of the coil.

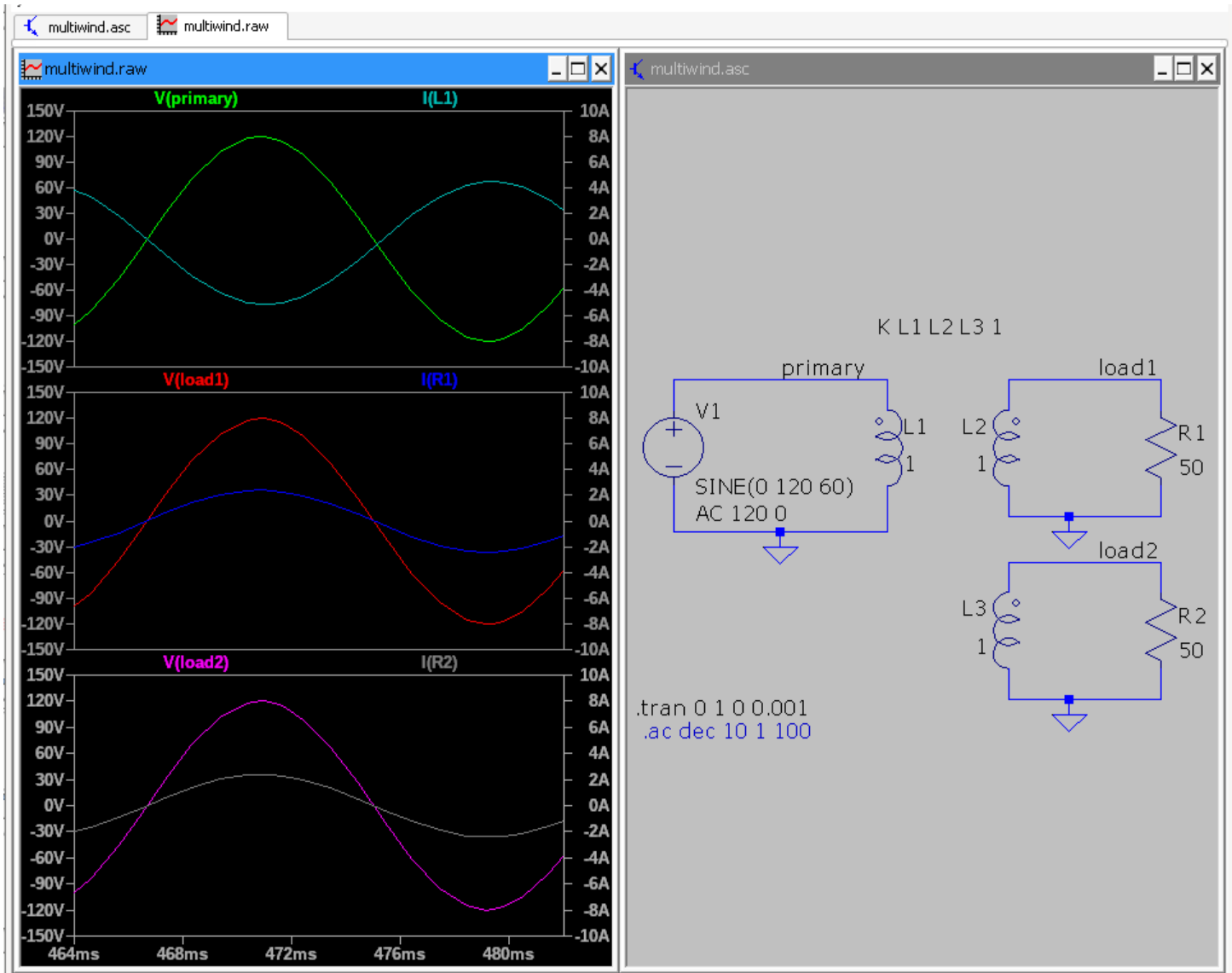


Figure 21: Transient Voltage, Current across Primary and Loads

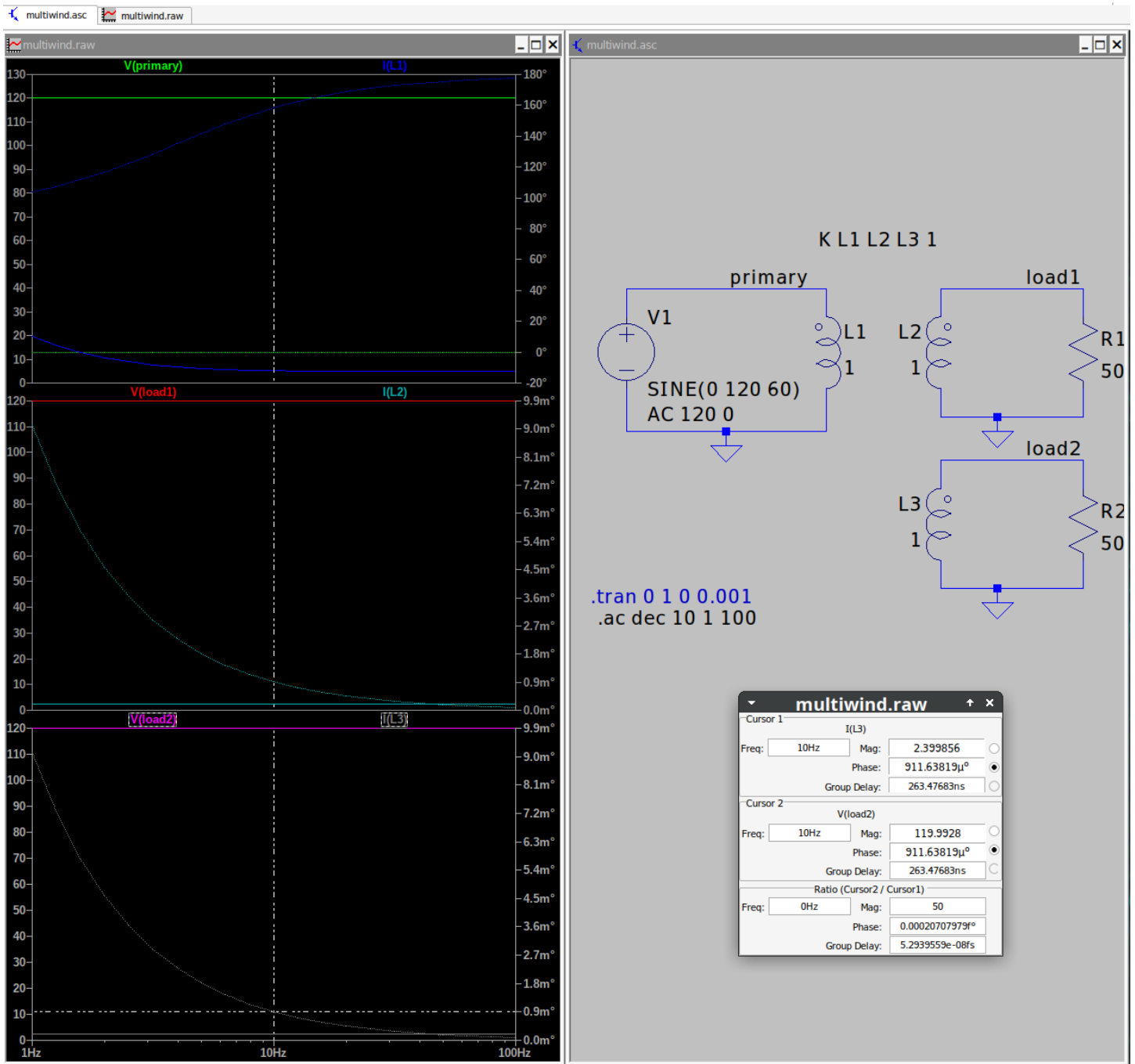


Figure 22: AC Voltage, Current across Primary and Loads

There is a slight voltage lead on the primary coil, and there is no lead or lag on the secondary coils.

The current convention also holds, similarly since the primary coil is a provider for the secondary coils, and Lenz + Faraday enforces the current to be 180 degrees out of phase from primary to secondary. Interestingly, we have multiple secondaries; we can see that the current waveforms across L2 and L3 are in-phase.

We observe the isolating effects. The magnitude of the voltage waveform between the primary coil and either load is the same, however, the primary current is split evenly between the two loads (since R1 is equal to R2).

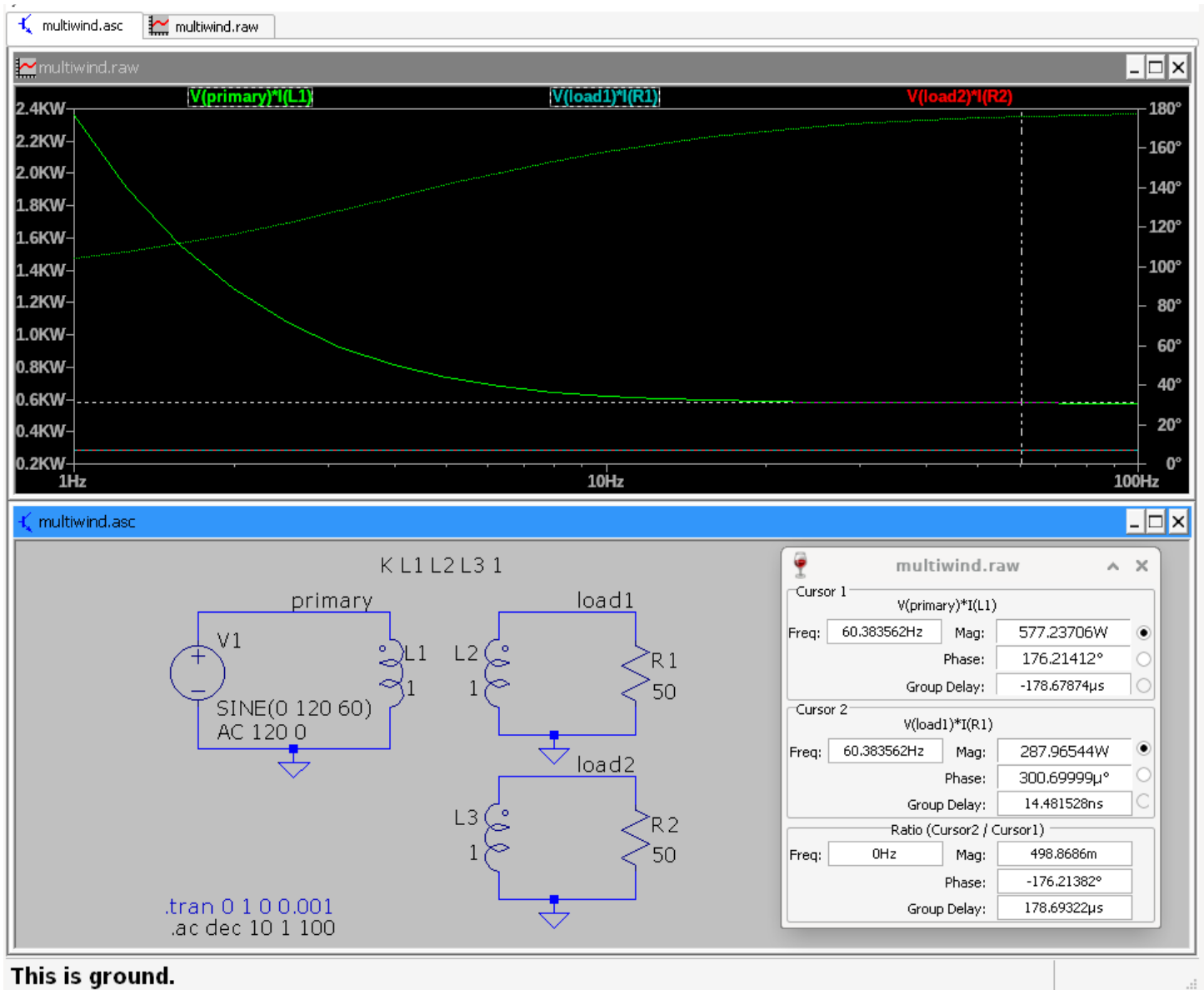


Figure 23: Observed power split between primary and each loads

We observe at 60Hz that our power is conserved, the power at the primary side is transferred without loss to our loads.

Feedback

This entire homework took me about 12-15 hours of work over 3 days, and I rewrote a couple of sections a couple of times. That could explain if some section doesn't make complete sense, or if i repeated myself at points.

The first question was ok, but the wording was a little vague at some points. For part C, I could not tell if i was supposed to have both sides match in current magnitude, since there was a note about observing the primary current rise to infinity.

The transformers were ok. Open ended is good, leaves a lot of room to play around (I went in circles at some points, such as figuring out whether to consider the phase dots).